

measure investment performance on a risk-adjusted basis. An annual alpha of 1% means your risk-adjusted excess return is 1% a year. Using CAPM, you could judge investment management skill just by tracking a portfolio's alpha. Economists also liked the fact that statistical significance could be associated with alpha and beta in the form of t-statistics and probability values.

They say there are two ways to gain a positive expected return. The first is by taking on known risk factors (beta). The second is by outsmarting everyone else (alpha).

There was just one problem with CAPM; it did not work very well in empirical testing. Returns of high beta portfolios were too low, and returns from low beta portfolios were too high. Much of the variation in expected return was unrelated to a portfolio's beta.

In the early 1980s, I came across an academic paper that tried to explain commodity returns in terms of single-factor CAPM. This paper was actually used in a graduate-level finance class at Berkeley. I wondered at the time, "What does the stock market have to do with the price of tea in China, or the price of any other commodity?" I was also aware at the time of the statistical problems associated with CAPM. Financial market returns, in general, violate the standard linear regression assumptions of drawing from distributions that are independent and identically distributed.⁶

CAPM ignored too many risks. Mark Twain reportedly said, "It ain't what we don't know that gets us in trouble. It's what we know for sure that just ain't so." Fischer Black said, "In the end, a theory is accepted not because it is confirmed by conventional empirical tests, but because researchers persuade one another that the theory is correct and relevant."

By the 1990s, academics had more evidence that all was not well with CAPM. Future returns on low P/E, low book-to-market, and small-cap stocks appeared to be higher than predicted by their betas. This resulted in the seminal Fama and French (1992) study that expanded CAPM from a single factor to a three-factor model by adding value and size risk factors to the single market factor. This well-known study suggested that the book-to-market ratio and market capitalization could explain the cross-sectional variation in average equity returns better than just the market factor alone. So now, we had three factors instead of just one. Not long afterward,